Structural Change Cheatsheet

Macroeconomics I - Exam Reference

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1 Notation and Definitions

Core Variables

Y_{it}	output in sector $i \in \{A, M, S\}$ (Agriculture, Manufacturing, Services)
K_{it}, L_{it}	capital and labor in sector i
$k_t = K_{1t}/K_t$	capital share in sector 1
$\lambda_t = L_{1t}/L_t$	labor share in sector 1
B_i	sector-specific TFP level
X_t	time-varying labor-augmenting productivity
γ_i	subsistence level for good i
η_i	weight in Stone-Geary utility
ϵ	substitution elasticity in CES aggregator
ω	weight in CES aggregator

Key Concepts

Concept	Definition
Structural Change	Reallocation of employment/output across sectors
Homothetic Preferences	Utility homogeneous of degree 1; linear Engel curves
Non-homothetic	Engel curves non-linear; income effects matter
Engel Curve	Demand as function of income: $x_i = x_i(I, p_i)$
Necessities	Income elasticity $\epsilon_I < 1$
Luxuries	Income elasticity $\epsilon_I > 1$
CGP	Constant Growth Path: aggregate grows at const. rate, sectors don't
Baumol's Cost Disease	Slow-growing sectors absorb labor as economy expands

2 Stylized Facts: Structural Change

Employment Shares Over Time (US)

Sector	Pattern	Key Observation
Agriculture	Sharp decline	From 70% (1800) to 2% (2000)
Manufacturing	Inverse-U shape	Peaks 30% mid-20th century, then declines
Services	Sharp rise	From 10% to 80%

Cross-Country Patterns

Relationship	Pattern	Interpretation
Agriculture share vs GDP	Negative	Rich countries have smaller Ag
Industry share vs GDP	Inverted-U	Peaks at middle income
Services share vs GDP	Positive	Rich countries have larger Services

Structural change is one of the most salient features of economic development that single-sector growth models cannot generate.

3 Demand-Side Explanations: Engel Curves

Homothetic vs Non-Homothetic Preferences

Type	Engel Curve	Implication for SC
Homothetic	Linear: doubling income $\Rightarrow$ doubling demand	No demand-driven SC
Necessities	Concave: $x' > 0$, $x'' < 0$	Declining share as income rises
Luxuries	Convex: $x' > 0$, $x'' > 0$	Rising share as income rises

Engel's Law

As income increases, the percentage spent on food (necessity) decreases, even though total food expenditure increases.

4 Stone-Geary Preferences

Utility Function

$$c_t = (c_t^A - \gamma^A)^{\eta_A} c_t^{M^{\eta_M}} (c_t^S + \gamma^S)^{\eta_S} \tag{Stone-Geary}$$

where $\gamma^A, \gamma^S > 0$ are subsistence levels and $\eta_A + \eta_M + \eta_S = 1$.

Special cases:

- $\gamma_i = 0, \forall i$: Cobb-Douglas (homothetic)
- $\gamma^A > 0$: food is a necessity
- $\gamma^S > 0$: services is a luxury (negative threshold)

Demand Derivation

Derivation of Stone-Geary Demand Household solves:

$$\max_{\{c_i\}} \mathcal{L} = \prod_i (c_i - \gamma_i)^{\eta_i} + \lambda \left[I - \sum_i p_i c_i \right]$$

FOC:

$$\eta_i (c_i - \gamma_i)^{-1} \prod_k (c_k - \gamma_k)^{\eta_k} = \lambda p_i$$

Dividing FOC for i by FOC for j:

$$\frac{\eta_i}{\eta_j} \frac{c_j - \gamma_j}{c_i - \gamma_i} = \frac{p_i}{p_j}$$

Solving the system:

$$c_j = \gamma_j + \frac{\eta_j (I - \sum_i p_i \gamma_i)}{p_j} \tag{Stone-Geary Demand}$$

$$\boxed{c_j = \gamma_j + \frac{\eta_j (I - \sum_i p_i \gamma_i)}{p_j}} \tag{Demand Schedule}$$

Key insight: Engel curve is linear but shifted by subsistence level. When income is low, only necessity goods are consumed.

5 Demand-Side Model: Production and Equilibrium

Sectoral Production Functions

$$Y_{it} = B_i F(K_{it}, X_t L_{it}) \quad i = \{A, M, S\} \tag{Sectoral Production}$$

where B_i is sector-specific TFP and X_t grows at rate g.

Market Clearing

$$\sum_i K_{it} = K_t, \quad \sum_i L_{it} = L_t \tag{Factor Market Clearing}$$

Key Assumption

Investment good is made from manufacturing good (no depreciation):

$$Y_{Mt} = \dot{K}_t + c_t^M L_t$$

Price Relations (Manufacturing = Numeraire)

$$p_t^A = \frac{B_M}{B_A}, \quad p_t^M = 1, \quad p_t^S = \frac{B_M}{B_S} \tag{Relative Prices}$$

Competitive Factor Markets

$$w_t = MPL_{it}, \quad r_t = MPK_{it} \quad \forall i \tag{Factor Price Equalization}$$

This implies:

$$\frac{K_{it}}{X_t L_{it}} = k_t \quad \forall i \tag{Common Capital-Output Ratio}$$

6 Constant Growth Path (CGP)

Definition

A **Constant Growth Path (CGP)** is when the consumption aggregate grows at a constant rate but sectoral consumptions do not.

Existence condition: CGP exists (unique) iff $\gamma^A/B_A = \gamma^S/B_S$.

Consumption Growth Rates on CGP

Derivation of Sectoral Growth Rates From intratemporal optimization:

$$\eta^M \frac{B_M(c_t^A - \gamma^A)}{B_A \eta^A} = c_t^M = \eta^M \frac{B_M(c_t^S + \gamma^S)}{B_S \eta^S}$$

With technology growth g (i.e., $X_t = X_0 e^{gt}$):

$$\frac{\dot{c}_t^A}{c_t^A} = g\frac{c_t^A - \gamma^A}{c_t^A} < g \qquad \text{(Agriculture)}$$

$$\frac{\dot{c}_t^M}{c_t^M} = g \qquad \text{(Manufacturing)}$$

$$\frac{\dot{c}_t^S}{c_t^S} = g\frac{c_t^S + \gamma^S}{c_t^S} > g \qquad \text{(Services)}$$

$$\boxed{\frac{\dot{c}^A}{c} < g < \frac{\dot{c}^S}{c} = g = \frac{\dot{c}^M}{c}} \qquad \text{(CGP Consumption Growth)}$$

As economy grows: agricultural consumption shrinks as share, manufacturing stays constant, services expand. This is Engel's Law in action.

7 Employment Dynamics (Demand Side)

Employment Growth Rates

Derivation of Employment Reallocation Using factor market clearing and production functions:

$$\frac{\dot{L}_{At}}{L_{At}} = n - g\frac{\gamma^A L_t}{B_A L_{At} X_t F(k^*, 1)} < n \qquad \text{(Agriculture)}$$

$$\frac{\dot{L}_{Mt}}{L_{Mt}} = n \qquad \text{(Manufacturing)}$$

$$\frac{\dot{L}_{St}}{L_{St}} = n + g\frac{\gamma^S L_t}{B_S L_{St} X_t F(k^*, 1)} > n \qquad \text{(Services)}$$

$$\boxed{\dot{L}_A/L_A < n < \dot{L}_S/L_S = n = \dot{L}_M/L_M} \qquad \text{(Employment Growth)}$$

Key Result: Demand-Driven SC

With Stone-Geary preferences (Engel's Law):

- Agricultural employment share declines
- Service employment share rises
- Manufacturing employment share stays constant (at CGP)

8 Supply-Side Explanations: Baumol's Cost Disease

Core Idea

If two sectors grow at different rates and share common production factors, there will be reallocation between them. There cannot be a BGP with constant sectoral shares.

Baumol's Cost Disease: Sectors with slower productivity growth see rising relative prices and absorb increasing labor shares because wages equalize across sectors.

Productivity Growth Differentials (Duarte & Restuccia)

Sector	Avg Growth Rate (1956-2004)
Agriculture	3.8%
Manufacturing	2.4%
Services	1.3%

9 Supply-Side Model: CES Aggregator

Final Good Aggregator (CES)

$$Y_t = \left[\omega Y_{1t}^{\frac{\epsilon-1}{\epsilon}} + (1-\omega) Y_{2t}^{\frac{\epsilon-1}{\epsilon}}\right]^{\frac{\epsilon}{\epsilon-1}} \tag{CES Aggregator}$$

where $\epsilon > 0$ is the substitution elasticity.

Sectoral Production

$$Y_{it} = A_{it} K_{it}^{\alpha_i} L_{it}^{1-\alpha_i} \quad i = 1, 2 \tag{Cobb-Douglas Sectors}$$

Assume $\alpha_1 < \alpha_2$: sector 1 is more labor-intensive.

Productivity Growth

$$\frac{\dot{A}_{it}}{A_{it}} = a_i > 0 \tag{Exogenous Tech Growth}$$

10 Supply-Side: Factor Allocation

Capital and Labor Shares

Define:

$$k_t \equiv \frac{K_{1t}}{K_t}, \quad \lambda_t \equiv \frac{L_{1t}}{L_t} \tag{Allocation Fractions}$$

Derivation of Capital Allocation From equalized MPKs:

$$\alpha_1 \omega Y_{1t}^{-\frac{1}{\epsilon}} \frac{Y_{1t}}{K_{1t}} = \alpha_2 (1-\omega) Y_{2t}^{-\frac{1}{\epsilon}} \frac{Y_{2t}}{K_{2t}}$$

Simplifying:

$$k_t = \left[1 + \frac{1-\omega}{\omega} \frac{\alpha_2}{\alpha_1} \left(\frac{Y_{2t}}{Y_{1t}} \right)^{\frac{\epsilon-1}{\epsilon}} \right]^{-1} \tag{Capital Share}$$

Similarly for labor:

$$\lambda_t = \left[1 + \frac{1-\alpha_2}{1-\alpha_1} \frac{\alpha_1}{\alpha_2} \frac{1-k_t}{k_t} \right]^{-1} \tag{Labor Share}$$

$$\boxed{k_t = \left[1 + \frac{1-\omega}{\omega} \frac{\alpha_2}{\alpha_1} \left(\frac{Y_{2t}}{Y_{1t}} \right)^{\frac{\epsilon-1}{\epsilon}} \right]^{-1}} \tag{Capital Allocation}$$

Key property: λ_t is increasing in k_t; labor and capital flow to the same sector.

11 Supply-Side: Response to Capital Accumulation

Capital Response to Economy Expansion

Derivation of Capital Reallocation Response Taking logs and differentiating k_t with respect to K_t:

$$\frac{\partial \log k_t}{\partial \log K_t} > 0 \iff (\alpha_2 - \alpha_1)(1 - \epsilon) > 0 \qquad \text{(Capital Response)}$$

This implies:

- If $\epsilon < 1$ (complements) and $\alpha_2 > \alpha_1$: capital flows to labor-intensive sector
- If $\epsilon > 1$ (substitutes) and $\alpha_2 > \alpha_1$: capital flows to capital-intensive sector

$$\boxed{\frac{\partial \log k_t}{\partial \log K_t} > 0 \iff (\alpha_2 - \alpha_1)(1 - \epsilon) > 0} \qquad \text{(Key Condition)}$$

Response to Technology Changes

$$\frac{\partial \log k_t}{\partial \log A_{2t}} = -\frac{\partial \log k_t}{\partial \log A_{1t}} > 0 \iff \epsilon < 1 \qquad \text{(Tech Response)}$$

When goods are complements ($\epsilon < 1$), faster technology growth in one sector causes capital to reallocate AWAY from that sector.

12 Supply-Side: Long-Run Dynamics

Growth and Reallocation

Derivation of Sectoral Growth Relation From MRP equalization in logs, taking time derivatives:

$$\frac{\epsilon - 1}{\epsilon}\gamma_{Y_1} - \gamma_{K_1} = \frac{\epsilon - 1}{\epsilon}\gamma_{Y_2} - \gamma_{K_2}$$

This implies:

$$\gamma_{K_1} - \gamma_{K_2} = \frac{\epsilon - 1}{\epsilon}(\gamma_{Y_1} - \gamma_{Y_2})$$

Therefore:

$$\text{sign}(\gamma_{K_1} - \gamma_{K_2}) = \text{sign}(\gamma_{Y_1} - \gamma_{Y_2}) \iff \epsilon > 1$$

$$\boxed{\begin{array}{lllll} \gamma_{L_1} > \gamma_{L_2} & \iff & \gamma_{K_1} > \gamma_{K_2} & \iff & \gamma_{Y_1} < \gamma_{Y_2} \quad \text{if } \epsilon < 1 \\ \gamma_{L_1} < \gamma_{L_2} & \iff & \gamma_{K_1} < \gamma_{K_2} & \iff & \gamma_{Y_1} < \gamma_{Y_2} \quad \text{if } \epsilon > 1 \end{array}} \qquad \text{(Sectoral Growth Relation)}$$

13 Long-Run Equilibrium

Factor Allocation in the Limit

Condition	Goods Complements ($\epsilon < 1$)	Goods Substitutes ($\epsilon > 1$)
Outcome	All factors go to slow-growing sector	All factors go to fast-growing sector
Growth rate	$\gamma = \min\{\gamma_{Y_1}, \gamma_{Y_2}\}$	$\gamma = \max\{\gamma_{Y_1}, \gamma_{Y_2}\}$
Mechanism	Rising relative price attracts labor	Productivity advantage dominates

Role of Investment Sector

If capital is made from one sector (e.g., manufacturing), in the limit:

- Employment goes to slowest-growing non-capital-goods sector
- Capital-goods sector survives due to its role in capital creation

14 Testing Demand vs Supply Explanations

Identification Problem

To distinguish demand-driven from supply-driven structural change, we need identifying moments:

Driver	Effect on Quantity	Effect on Price
Demand increase	$\uparrow$	$\uparrow$
Productivity increase	$\uparrow$	$\downarrow$

With both quantity and price data, we can distinguish demand from supply shocks. For structural change: changes in employment associated with income changes (demand) vs relative price changes (supply).

15 Empirical Findings

Key Papers and Conclusions

Paper	Method	Conclusion
Buera & Kaboski (2009)	Stone-Geary + supply-side	Cannot match steep decline in manufacturing; needs low ϵ
Herrendorf et al. (2013)	Non-homothetic CES	Both channels matter; final exp. vs value-added matters
Boppart (2014)	PIGL preferences	Income effects don't vanish; 50/50 split
Comin et al. (2021)	Non-homothetic CES	Non-homotheticities explain 3/4 of reallocation

Parameter Estimates (Herrendorf et al.)

Parameter	Final Expenditure	Value Added
σ	0.85	0.002 (Leontief)
$\overline{c}_A$	-1,350	-139
$\overline{c}_S$	11,237	4,262

16 Problem-Solving Templates

(a) Derive Demand from Stone-Geary Preferences

1. Set up Lagrangian: $\mathcal{L} = \prod_i (c_i - \gamma_i)^{\eta_i} + \lambda(I - \sum_i p_i c_i)$
2. Take FOCs: $\eta_i (c_i - \gamma_i)^{-1} \prod_k (c_k - \gamma_k)^{\eta_k} = \lambda p_i$
3. Divide FOC i by FOC j to get ratio form
4. Solve system for c_i in terms of I and p_i
5. Result: $c_i = \gamma_i + \frac{\eta_i (I - \sum_j p_j \gamma_j)}{p_i}$

(b) Analyze CGP in Demand-Side Model

1. Check CGP existence: $\gamma^A / B_A = \gamma^S / B_S$
2. From intratemporal FOCs, derive consumption ratios
3. Use technology growth rate g to find consumption growth rates
4. Result: $\dot{c}^A / c < g < \dot{c}^S / c$
5. Derive employment growth: $\dot{L}_A / L < n < \dot{L}_S / L$

(c) Analyze Supply-Side Reallocation

1. Set up CES aggregator and sectoral production functions
2. Derive factor allocation: k_t and λ_t
3. Compute responses to K and A changes
4. Apply condition: $(\alpha_2 - \alpha_1)(1 - \epsilon) > 0$
5. Determine long-run: complements $\Rightarrow$ slow sector wins

(d) Compare Demand vs Supply Explanations

1. Identify key predictions of each theory
2. Demand: Engel curves $\Rightarrow$ consumption shares shift with income
3. Supply: productivity differentials $\Rightarrow$ relative price changes
4. Find identifying moment: prices vs quantities
5. Use data to assess relative importance

17 Common Pitfalls

1. **Confusing homothetic with non-homothetic**
   - Homothetic: linear Engel curves, no demand-driven SC
   - Non-homothetic: non-linear Engel curves, demand-driven SC possible
2. **Forgetting CGP existence condition**
   - CGP exists iff $\gamma^A/B_A = \gamma^S/B_S$
   - Without this, sectoral growth rates are not constant
3. **Misinterpreting complementarity**
   - $\epsilon < 1$: goods are complements; faster tech $\Rightarrow$ reallocation AWAY
   - $\epsilon > 1$: goods are substitutes; faster tech $\Rightarrow$ reallocation TOWARD
4. **Confusing sectoral growth with reallocation**
   - Fast-growing sector can still see employment decline if reallocation is strong enough
5. **Ignoring the role of the investment sector**
   - Sector that produces capital goods survives even if slow-growing
6. **Misinterpreting Engel curves**
   - Necessity: share declines but absolute consumption rises
   - Luxury: share rises and absolute consumption rises faster
7. **Confusing final expenditure with value-added**
   - These give very different parameter estimates
   - Affects conclusions about demand vs supply importance

18 Quick Reference Formulas

Stone-Geary Preferences

$$c_t = (c_t^A - \gamma^A)^{\eta_A} c_t^{M^{\eta_M}} (c_t^S + \gamma^S)^{\eta_S}$$
$$c_j = \gamma_j + \frac{\eta_j (I - \sum_i p_i \gamma_i)}{p_j}$$

Production

$$Y_{it} = B_i F(K_{it}, X_t L_{it})$$
$$Y_t = [\omega Y_{1t}^{\frac{\epsilon-1}{\epsilon}} + (1-\omega) Y_{2t}^{\frac{\epsilon-1}{\epsilon}}]^{\frac{\epsilon}{\epsilon-1}}$$

CGP Growth Rates (Demand Side)

$$\frac{\dot{c}^A}{c} = g \frac{c^A - \gamma^A}{c^A} < g$$
$$\frac{\dot{c}^M}{c} = g$$
$$\frac{\dot{c}^S}{c} = g \frac{c^S + \gamma^S}{c^S} > g$$

Employment Growth (Demand Side)

$$\frac{\dot{L}_A}{L} < n$$
$$\frac{\dot{L}_M}{L} = n$$
$$\frac{\dot{L}_S}{L} > n$$

Capital Allocation (Supply Side)

$$k_t = \left[1 + \frac{1-\omega}{\omega} \frac{\alpha_2}{\alpha_1} \left(\frac{Y_{2t}}{Y_{1t}} \right)^{\frac{\epsilon-1}{\epsilon}} \right]^{-1}$$
$$\frac{\partial \log k}{\partial \log K} > 0 \iff (\alpha_2 - \alpha_1)(1-\epsilon) > 0$$

Key Conditions

- CGP exists: $\gamma^A / B_A = \gamma^S / B_S$
- Capital response: $(\alpha_2 - \alpha_1)(1-\epsilon) > 0$
- Long-run: $\epsilon < 1 \Rightarrow$ slow sector wins; $\epsilon > 1 \Rightarrow$ fast sector wins

19 Implementation Notes

Source Material

- Lecture 6: Structural Change
- Key Focuses: Multi-sector models, demand/supply explanations, Engel curves

Structure

Following the Quality Ladder Cheatsheet style with:

- Derivation boxes for step-by-step derivations
- Result boxes for key formulas
- Intuition boxes for economic interpretations
- Warning boxes for common mistakes
- Tables for comparisons

Packages Used

- amsmath, amssymb, mathtools: Mathematics
- graphicx: Graphics
- xcolor: Color support
- siunitx: Number formatting
- cleveref: Cross-references
- hyperref: Hyperlinks
- tcolorbox: Colored boxes
- booktabs: Tables

Compilation

Compile with: `pdflatex` or `lualatex`